Outlook

ATiGB2026 ACCEPTED NOTIFICATION FOR PAPER ID: 39

From ATiGB2026 <atigb2026@easychair.org>
Date Tue 5/12/2026 12:51 PM
To Tai Vo <S4140277@student.rmit.edu.au>

Dear Chau Le, Anh Pham, Mai Nguyen, Nam Pham, Minh-Tai Vo,

We are pleased to inform you that your paper ID 39, entitled "Obstacle Avoidance for Collaborative Robots Using Modulation Dynamical Systems Technique" has been ACCEPTED for presentation at the ATiGB2026 Conference.

Please check these instructions in order to clarify the next steps to follow:

1. Authors are requested to revise their manuscript by incorporating the reviewers' comments and addressing the similarity report. The updated version must adhere to the same template provided on the conference website.
- Provide a point-by-point response to the reviewers' comments and include it on the last page of the revised paper.
- Clearly highlight all modifications made in the revised manuscript.
- Ensure that the similarity report is less than 20%.

2. Authors upload the REVISED VERSION and the response file into the submission system before 24/05/2026.

3. Please registration at link: https://aus01.safelinks.protection.outlook.com/?url=https%3A%2F%2Fforms.gle%2FjBRzctaAGW7CPZ2H6&data=05%7C02%7CS4140277%40student.rmit.edu.au%7Cdc5bdefe1d0f488eec7208deafea73d1%7Cd1323671cdbe4417b4d4bdb24b51316b%7C0%7C0%7C639141618701149444%7CUnknown%7CTWFpbGZsb3d8eyJFbXB0eU1hcGkiOnRydWUsIlYiOiIwLjAuMDAwMCIsIlAiOiJXaW4zMiIsIkFOIjoiTWFpbCIsIldUIjoyfQ%3D%3D%7C0%7C%7C%7C&sdata=d8ye%2BFjGq9eehXKYLTAbBlGQ9fZG%2F2PBBrvjF7VkFII%3D&reserved=0

4. After registration is adequately completed, the invitation letter will be sent to the authors.

Please do not hesitate to contact us again at atigb@ute.udn.vn if you need further information.

Thank you for contributing to the ATiGB2026; we hope to see you in Danang.
With kind regards,
Minh Tien Nguyen
ATiGB2026 Organizing Committee

SUBMISSION: 39
TITLE: Obstacle Avoidance for Collaborative Robots Using Modulation Dynamical Systems Technique

---------- REVIEW 1 ----------

SUBMISSION: 39
TITLE: Obstacle Avoidance for Collaborative Robots Using Modulation Dynamical Systems Technique

---------- Overall evaluation ----------

SCORE: 5 (Accept with revisions)
----- TEXT:
The paper presents a well-structured and logical simulation study on applying the MDS method for UR3 cobots. The mathematical foundation is sound, and the results successfully demonstrate the algorithm's potential for real-time reactive control.

Therefore, I recommend Accept with revisions. I believe the manuscript is suitable for publication after the authors address a few key issues to strengthen their work:

1. Including simulation scenarios for ellipsoidal and moving obstacles to fully validate the theoretical claims.

2. Adding a brief comparison with traditional methods (e.g., APF) to highlight the proposed method's advantages.
---------- Important of the topic ----------
SCORE: 3 (fair)
---------- Quality of Content ----------
SCORE: 3 (fair)
---------- Clarity and Organization ----------
SCORE: 4 (good)
---------- Format, References and Citations ----------
SCORE: 3 (fair)
---------- Reviewer's comments ----------
There are several issues that need to be addressed:

1. The authors introduced the mathematical model for an ellipsoidal obstacle in Eq. (2) , but the simulation results explicitly demonstrate only cylindrical or spherical obstacles. The authors should clarify the parameters of the obstacles in Section IV or add a specific simulation scenario with a non-uniform ellipsoid ($a \neq b \neq c$) to fully validate the theoretical model.

2. The introduction mentions navigating dynamic environments with moving obstacles. Furthermore, the MDS approach is theoretically well-suited for real-time dynamic obstacle avoidance (as seen in Ref [10]). However, the simulation scenarios in Section IV only evaluate static obstacles. To strengthen the paper, the authors should include a simulation scenario featuring a moving obstacle to demonstrate the real-time reactive capabilities of implemented algorithm.

3. The APF method has seen many improvements to fix the local minima problem. The authors should include APF simulation results as a comparison to clearly highlight the advantages of the proposed method.